

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 — Scenario-Based Requirement Analysis

Course Code:	CSA1007	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 1 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills.</i>		
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Date:	28/07/2026	Duration:	60 minutes

Code of Conduct

I V.Rakshana(192425137), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V. Rakshana

Problem Statement 24: Smart Museum Artifact Preservation Platform

1. Scenario Study and Analysis

1.1 Business Domain

The assigned problem statement belongs to the Museum Management and Cultural Heritage Preservation domain. Museums preserve historical artifacts, paintings, sculptures, manuscripts, fossils, and cultural objects for future generations. Traditional museum management methods often rely on manual inspection, paper-based documentation, and periodic maintenance schedules, making it difficult to continuously monitor artifact conditions. The platform helps museums improve artifact preservation while ensuring better visitor engagement and operational efficiency.

1.2 Problem Description

Design and develop a Smart Museum Artifact Preservation Platform that continuously monitors environmental conditions such as temperature, humidity, lighting, and air quality using IoT sensors. The platform predicts artifact deterioration using AI models, generates preservation recommendations, schedules maintenance activities, tracks restoration history, manages digital inventories, and provides administrators with dashboards and reports for effective museum management.

1.3 Existing Challenges

- Manual monitoring of artifacts increases the possibility of human error.
- Environmental conditions are not continuously monitored.
- Artifact deterioration is identified only after visible damage occurs.
- Maintenance schedules are manually maintained.
- Digital records are incomplete or inconsistent.

1.4 System Objectives

- Monitor environmental conditions continuously.
- Detect early signs of artifact deterioration using AI.
- Maintain complete digital records for every artifact.
- Schedule preventive maintenance automatically.
- Generate preservation recommendations.

1.5 Key Stakeholders

Stakeholder	Role	Primary Interest
Museum Director	Overall management	Preservation efficiency
Curator	Artifact management	Artifact safety
Conservation Specialist	Restoration	Early damage detection

Stakeholder	Role	Primary Interest
Museum Staff	Daily monitoring	Easy management

2. Functional and Non-Functional Requirements 2.1

Functional Requirements

ID	Module	Requirement
FR-1	Artifact Management	Register new artifacts
FR-2	Artifact Management	Update artifact information
FR-3	Environmental Monitoring	Collect temperature and humidity
FR-4	Environmental Monitoring	Monitor air quality
FR-5	AI Prediction	Predict deterioration risks

2.2 Non-Functional Requirements

Quality Attribute	Requirement
Performance	Dashboard loads within 3 seconds
Quality Attribute	Requirement

Scalability	Supports multiple museums
Security	Role-based authentication and encryption
Reliability	99.5% uptime
Availability	Continuous monitoring

3. Actors and Use Cases

Actor	Type	Use Cases
Museum Director	Primary	View reports, Dashboard
Curator	Primary	Register artifacts, Update details
Conservation Specialist	Primary	View alerts, Schedule maintenance
Museum Staff	Primary	Monitor artifacts
Visitors	Secondary	View digital information

4. Software Requirement Specification (SRS)

4.1 Introduction and Scope

This Software Requirement Specification describes the Smart Museum Artifact Preservation Platform. The system integrates IoT sensors, Artificial Intelligence, cloud databases, and web technologies to preserve museum artifacts through continuous monitoring and predictive maintenance.

In Scope

- Artifact registration
- Environmental monitoring
- AI prediction
- Maintenance scheduling

4.3 Data Requirements

Artifact	ID, Name, Category, Age, Material
Sensor	Temperature, Humidity, Light
Museum Room	Room ID, Location
Maintenance	Date, Staff, Status
Restoration	Details, Cost

4.5 System Features

- AI-based deterioration prediction

- IoT environmental monitoring
- Digital artifact inventory
- Automatic maintenance scheduling
- Dashboard visualization

4.6 Deployment Considerations

The platform should be deployed using cloud infrastructure with scalable services. IoT sensors installed throughout museum galleries continuously collect environmental data. AI services process incoming sensor data to identify deterioration risks. Administrators access the system through secure web dashboards while mobile applications provide maintenance staff with instant notifications.

5. Assumptions and Constraints

5.1 Assumptions

- IoT sensors are installed properly.
- Internet connectivity is available.
- Museum staff are trained.
- Historical artifact data is available.
- AI models are periodically updated.

6. Requirement Validation

6.1 Completeness Check

All stakeholders have corresponding functional requirements. Museum administrators receive reporting capabilities, conservation specialists receive maintenance scheduling, curators manage artifacts, AI predicts deterioration, and IoT devices continuously monitor environmental conditions.

6.2 Consistency and Ambiguity Check

- Environmental monitoring intervals are clearly defined.
- AI prediction is based on historical and sensor data.
- Maintenance scheduling follows administrator approval.
- No conflicting functional requirements were identified.

6.3 Suggested Improvements

- Add Digital Twin visualization.
- Support Augmented Reality for visitors.
- Introduce blockchain for artifact history.
- Implement multilingual support.

7. Conclusion

The Smart Museum Artifact Preservation Platform provides an intelligent solution for preserving valuable museum collections using Artificial Intelligence, IoT, and cloud technologies. The system automates artifact monitoring, predicts deterioration risks, schedules maintenance, maintains digital inventories, and generates comprehensive reports. This requirement analysis establishes a solid foundation for the design, development, and deployment of a scalable, secure, and efficient smart museum preservation system.